

Homework 6

Please give your final answer as a number where possible as well as show work for full credit.

Question 1. Please read chapter 9 of Chartrand et al. and write a couple sentences about a topic/example/concept that you found difficult or interesting and why?

Every concept after Q6 confused me. I dont really understand how generating functions work intuitively.

Question 2. What is the exact term in $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^{10}$ containing x ?

$$\begin{aligned} \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^{10} &= \sum_{k=0}^{10} \binom{10}{k} (\sqrt{x})^{10-k} \left(\frac{-1}{\sqrt{x}}\right)^k \\ x^{(10-k)/2} (-x^{-k/2}) &= x^{\frac{10-k-k}{2}} (-1)^k = x^{5-k} (-1)^k \\ \sum_{k=0}^{10} \binom{10}{k} x^{5-k} (-1)^k & \\ \text{When } k=4, \binom{10}{4} &= 210, \quad x^{5-4} = x^1 = x \end{aligned}$$

Question 3. Use the Binomial Theorem to simplify the sum,

$$\begin{aligned} &2^{n^2} + \binom{n}{1} 2^{n(n-1)} + \binom{n}{2} 2^{n(n-2)} + \cdots + \binom{n}{n-1} 2^n + 1 \\ &= \sum_{k=0}^n \binom{n}{k} (2^n)^{n-k} 1^k = (2^n + 1)^n \end{aligned}$$

Question 4. How many different 8-digit numbers can be obtained by permuting the digits in the number 70, 440, 704? (Since each number is an 8-digit number, the first digit cannot be 0.)

$$\frac{8!}{3!3!2!} - \frac{7!}{3!2!2!} = 350$$

Question 5. A professor has 10 identical new pens that he no longer needs. In how many ways can these pens be given to 3 students if

- (a) there are no other conditions?
- (b) every student must receive at least one pen?
- (c) every student must receive at least two pens?
- (d) every student must receive at least three pens?

$$\begin{aligned} \text{(a)} \quad &\binom{10+3-1}{3-1} = \binom{12}{2} = 66 \\ \text{(b)} \quad &\binom{10-3+3-1}{3-1} = \binom{9}{2} = 36 \\ \text{(c)} \quad &\binom{10-6+3-1}{3-1} = \binom{6}{2} = 15 \\ \text{(d)} \quad &\binom{10-9+3-1}{3-1} = \binom{3}{2} = 3 \end{aligned}$$

Question 6. How many 4-digit numbers are there, the sum of whose digits is 11?

Find cases where $a > 0$:

$$\begin{aligned} a + b + c + d &= 11 \\ x &= a - 1 \\ x + 1 + b + c + d &= 11 \implies x + b + c + d = 10 \\ \binom{10+4-1}{4-1} &= 286 \quad // \text{ count for } a > 0 \end{aligned}$$

$$\begin{aligned} &\text{Remove cases where } a > 9: \text{ If } a > 9, \quad y = a - 10 \quad (\text{aka } y = x - 9), \\ y + 10 + b + c + d &= 11 \implies y + b + c + d = 1, \\ \binom{1+4-1}{4-1} &= 4. \end{aligned}$$

$$\begin{aligned} &\text{Remove cases where } b, c, d > 9: \text{ For } b > 9, \quad y = b - 10 \\ x + y + 10 + c + d &= 10 \implies x + y + c + d = 0 \end{aligned}$$

$$\binom{0+4-1}{4-1} = 1$$

Repeat the above for c and d to get 1 and 1

$$\binom{13}{3} - 4 - 1 - 1 - 1 = 279$$

Question 7. A man enters Ben's Bagels to buy a dozen bagels only to learn that even though Ben has a large supply of plain bagels and blueberry bagels, he only has two cinnamon bagels left and no other kinds.

- Express the number of selections of r bagels ($r \geq 0$) the man can make as a product of polynomials and/or power series¹.
- Determine the coefficient of x^{12} of the product in (a). What information does this coefficient provide?
- Since plain and blueberry supplies are infinite, we can state them as $\frac{1}{1-x}$
Cinnamon bagels are limited to 2, so we can state them as $\sum_{k=0}^{n=2} (x^k) = 1 + x + x^2$
 $(1 + x + x^2)(\frac{1}{1-x})^2 = \frac{1+x+x^2}{(1-x)^2}$
- $\frac{1+x+x^2}{1-x^2} = (1 + x + x^2)(1 - x)^{-2}$
We know that we can expand $(1 - x)^{-2} = 1 + 2x + 3x^2 + \dots + (n + 1)x^n + \dots$
We can see that we can find the coefficient with $1 + n$
If we distribute, we get $1(1 - x)^{-2} + x(1 - x)^{-2} + x^2(1 - x)^{-2}$
That simplifies to $(n + 1) + (n) + (n - 1)$ when we look just for the coefficient since x and x^2 shifts the expansion
Since $n = 12$, we get $13 + 12 + 11 = 36$
This number tells us the number of ways we can get 12 bagels out of all our choices. If we plugged in 13 then it would be for 13 bagels, etc.

Question 8. Suppose that we are interested in the number of non-negative integer solutions a, b and c of the equation $a + 2b + 3c = r$, where $r \in \mathbb{N}$.

- What is this number when $r \in \{1, 2, 3, 4\}$?
- Use generating functions to describe this number for a general $r \in \mathbb{N}$.
- $r=1$: 1 solution
 - $r=2$: 2 solutions
 - $r=3$: 3 solutions
 - $r=4$: 4 solutions
- the max value of c is $r/3$ since $a + 2b$ can be written as $r - 3c$
 $\max b = \frac{r-3c}{2}$
 a cannot be 0 so we must add 1
since c is the largest contributor, $n = r/3$
so, solutions = $\sum_{c=0}^{n=r/3} (\frac{r-3c}{2} + 1)$

Question 9. A man buys 6 doughnuts, each of which is a plain doughnut, a powdered doughnut or a glazed doughnut. How many possible selections are there if he buys at least one doughnut of each kind?

- Answer the question above by listing all possible selections in a table.
- Answer the question above by determining the coefficient of x^6 in a product of polynomials and/or power series.
- Answer the question above by computing $\binom{s+t-1}{s}$ for an appropriate choice of s and t .

¹The book has a list of common power series in figure 9.8 on page 347.

1	1	4
1	2	3
1	3	2
1	4	1
2	1	3
2	2	2
2	3	1
3	1	2
3	2	1
4	1	1

TABLE 1. Selections Table.

- (a)
- (b) we rewrite the selections for the donuts each as $x/(1-x)$.
for all three types of donuts, we get $(\frac{x}{1-x})^3 = \frac{x^3}{(1-x)^3} = x^3(1-x)^{-3}$
expanding $(1-x)^{-k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n$
we let $k = 3$ since we're trying to find x^6 and we have $(1-x)^{-3}$
plugging in, we get $\sum_{n=0}^{\infty} \binom{n+3-1}{3-1} x^n = \sum_{n=0}^{\infty} \binom{n+2}{2} x^n$
 $\binom{5}{2} = 10$
- (c) we can set s = number of required donuts since there must be at least 1 type of each. t can be the extra 3 that can be anything which comes out $\binom{3+3-1=5}{3} = 10$